# Aoxuan (Douglas) Li

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## Education

**PhD in Computer Science**
IMDEA Software Institute, Dec 2024 – Present

**Master of Science in Computer Science**
University of California, Los Angeles (UCLA), Sep 2017 – Jun 2019
*GPA: 3.81/4.0 | Master Thesis: On Explicit Depth Robust Graphs*

**Bachelor of Science in Computing**
Macao Polytechnic Institute, 2013 – 2017
*Overall GPA: 3.56/4.0*

## Experience

**Cryptographer | ExpandZK (Mystiko.Network)**
Aug 2021 – Dec 2024 (Freelance) | Nov 2020 – Aug 2021

- Designed the core protocol for **Mystiko.Network**, which achieved over **$280M in total trading volume** and $1M transactions within its first year.
- Architected the decentralized auditing committee mechanism and optimized ZK-Rollup batching to reduce on-chain data costs by up to **90%**.

**Researcher | Macao Polytechnic University, Macao & University of Bologna, Italy**
Aug 2021 – Aug 2024

- Conducted blockchain simulation research on ICP architecture advised by Prof. Gabriele D'Angelo.
- Researched distributed ledger technologies, DeFi confidentiality, and smart contract security.

**Cryptography Researcher (Intern) | Nethermind**
Aug 2023 – Nov 2023

- Optimized STARK-based proof systems, reducing final argument proof size and verification complexity.

**Software Engineer | Stealth Software Technologies, Los Angeles**
July 2019 – July 2020

- Contributed to **PULSAR/VLDS** (Verifiable Lean Database Systems) and **SAFRN** (Secure Architecture for Resilient Networks).
- Designed a novel Private Set Intersection (PSI) algorithm and implemented cryptographic primitives in Java.

**Graduate Research Assistant | UCLA (Cryptography Group)**
Jan 2018 – June 2019

- Conducted theoretical cryptography research advised by **Prof. Rafail Ostrovsky**.
- Focused on the design of explicit depth-robust graphs and cryptographic protocol analysis.

**Graduate Research Assistant| UCLA (Cuneiform Digital Library Initiative)**
Jan 2018 – June 2019

- Managed large-scale CentOS and MySQL server infrastructure for global digital heritage data.
- Optimized backup strategies for Docker-based virtualization and developed a CakePHP-based framework for data storage and display.

## Selected Publications

Aoxuan Li, Gabriele D'Angelo, Su-Kit Tang, Frank Fang, and Baron Gong. An auditable confidentiality protocol for blockchain transactions. *Blockchain: Research and Applications*, 2025

AoXuan Li, Luca Serena, Mirko Zichichi, Su-Kit Tang, Gabriele D'Angelo, and Stefano Ferretti. Modelling of the internet computer protocol architecture: The next generation blockchain. In *International Congress on Blockchain and Applications*, pages 3–12. Springer International Publishing Cham, 2022

AoXuan Li, Su-Kit Tang, and Gabriele D'Angelo. Implementation and preliminary evaluation of an auditable confidentiality mechanism for defi. In *2023 IEEE 43rd International Conference on Distributed Computing Systems Workshops (ICDCSW)*, pages 49–54. IEEE, 2023

Luca Serena, AoXuan Li, Mirko Zichichi, Gabriele D'Angelo, Stefano Ferretti, and Su-Kit Tang. Simulation of the internet computer protocol: the next generation multi-blockchain architecture. In *2022 IEEE/ACM 26th International Symposium on Distributed Simulation and Real Time Applications (DS-RT)*, pages 119–126. IEEE, 2022

Kaiyuan Tang, AoXuan Li, and Su-Kit Tang. Fully on-chain cloud storage dapp on the internet computer protocol. In *2023 IEEE 43rd International Conference on Distributed Computing Systems Workshops (ICDCSW)*, pages 43–48. IEEE, 2023

Aoxuan Li. On explicit depth robust graphs. Master's thesis, University of California, Los Angeles, 2019

## Technical Skills & Service

**Skills:** Python, Java, LaTeX, SQL, Docker.

**Academic Service:** Subreviewer for **Eurocrypt 2026/2019**, **CHES 2026**, and **INFOCOM 2023**.

**Teaching:** TA for Security and Cryptography and Optimization Methods at MPU.

**Languages:** Mandarin (Native), English (Fluent), Cantonese (Basic).